

Data Dictionary — Entity Relationship Diagram

E-Commerce Platform | System Documentation

User

Stores all registered user accounts, credentials, and embedded shipping addresses.

Field	Type	Constraint	Description
_id	ObjectId	PK	Auto-generated MongoDB document identifier
name	String	Required	Full display name of the user
email	String	Required, Unique	Login email address
password	String	Required	bcrypt-hashed password (never stored in plain text)
role	String	Required	Enum: user admin
isVerified	Boolean	Optional	True once the user confirms their email
addresses	Array	Optional	Embedded address objects: {street, city, postalCode, country, isDefault}
passwordResetToken	String	Optional	SHA-256 hashed token for password reset flow
passwordResetExpires	Date	Optional	Expiry timestamp for the reset token
createdAt	Date	Auto	Account creation timestamp (Mongoose timestamps)
updatedAt	Date	Auto	Last modification timestamp (Mongoose timestamps)

Product

Catalogue of all products available for purchase.

Field	Type	Constraint	Description
_id	ObjectId	PK	Auto-generated document identifier
name	String	Required	Product display name
slug	String	Unique	URL-friendly identifier, auto-derived from name
description	String	Required	Full product description
shortDescription	String	Optional	Brief summary shown on product cards
price	Number	Required	Current selling price in base currency
comparePrice	Number	Optional	Original price shown as strikethrough
stock	Number	Required	Available inventory count
status	String	Required	Enum: active inactive draft
category	ObjectId	FK → Category	Primary product category reference
images	Array	Required	Cloudinary image objects: [{url, publicId}]
averageRating	Number	Optional	Computed average of all review ratings (0-5)
reviewCount	Number	Optional	Total number of reviews for this product
tags	Array<String>	Optional	Keywords for search and filtering

Category

Hierarchical product categories. A category may have a parent for sub-categories.

Field	Type	Constraint	Description
_id	ObjectId	PK	Auto-generated document identifier
name	String	Required, Unique	Category display name
slug	String	Unique	URL-friendly identifier
description	String	Optional	Category description shown on listing pages

image	String	Optional	Banner or icon URL for the category
parent	ObjectId	FK → Category, Optional	Reference to parent category (enables hierarchy)

Order

Records a customer purchase from placement through fulfilment.

Field	Type	Constraint	Description
_id	ObjectId	PK	Auto-generated document identifier
user	ObjectId	FK → User, Req	Reference to the ordering customer
orderNumber	String	Unique	Human-readable reference (e.g. ORD-00045)
items	Array	Required	Line items: [{product, name, price, qty, image}]
shippingAddress	Object	Required	Address snapshot captured at checkout time
status	String	Required	Enum: pending confirmed processing shipped delivered cancelled refunded
paymentMethod	String	Required	Enum: card mtn_momo airtel pesapal paypal stripe
paymentStatus	String	Required	Enum: pending paid failed refunded
subtotal	Number	Required	Sum of item prices before discount and shipping
discount	Number	Optional	Discount amount from applied coupon
shippingCost	Number	Optional	Shipping fee added to the total
total	Number	Required	Final amount charged to the customer
couponCode	String	Optional	Coupon code applied at checkout

Cart

Persisted shopping cart — one cart per registered user.

Field	Type	Constraint	Description
_id	ObjectId	PK	Auto-generated document identifier
user	ObjectId	FK → User, Unique	Cart owner; unique index ensures one cart per user
items	Array	Optional	Cart line items: [{product, name, price, qty, image}]
total	Number	Optional	Computed sum of all item prices
couponCode	String	Optional	Coupon code applied to the cart
discount	Number	Optional	Discount amount from the applied coupon

Payment

Transaction record for every payment attempt across all gateways.

Field	Type	Constraint	Description
_id	ObjectId	PK	Auto-generated document identifier
order	ObjectId	FK → Order, Req	Order this payment is associated with
user	ObjectId	FK → User, Req	Customer who made the payment
method	String	Required	Enum: stripe paypal mtn_momo airtel pesapal
gateway	String	Optional	Actual gateway used (e.g. pesapal aggregating MTN MoMo)
status	String	Required	Enum: pending processing completed failed refunded
amount	Number	Required	Amount charged
currency	String	Required	ISO 4217 code: UGX, USD, etc.
transactionId	String	Optional	Gateway-issued transaction reference
gatewayResponse	Mixed	Optional	Raw webhook or callback payload stored for audit

Review

Customer reviews and star ratings attached to products.

Field	Type	Constraint	Description
_id	ObjectId	PK	Auto-generated document identifier
user	ObjectId	FK → User, Required	Review author
product	ObjectId	FK → Product, Req	Reviewed product
rating	Number	Required	Star rating 1–5
title	String	Optional	Review headline
body	String	Required	Full review text
verifiedPurchase	Boolean	Optional	True if reviewer has a completed order for this product
createdAt	Date	Auto	Submission timestamp

Wishlist

Products saved by a user for later viewing — one wishlist per user.

Field	Type	Constraint	Description
_id	ObjectId	PK	Auto-generated document identifier
user	ObjectId	FK → User, Unique	Wishlist owner; unique index ensures one per user
items	Array	Optional	Saved products: [{product (ObjectId), addedAt (Date)}]

UserActivity

Behavioural event log used for analytics and AI recommendation model training.

Field	Type	Constraint	Description
_id	ObjectId	PK	Auto-generated document identifier
user	ObjectId	FK → User, Required	User who performed the action
product	ObjectId	FK → Product, Optional	Product involved in the event
activityType	String	Required	Enum: view search cart_add cart_remove wishlist_add purchase review rec_impression similar_impression
metadata	Mixed	Optional	Free-form context: search query, referrer URL, recommendation type, etc.
timestamp	Date	Required	Event time; indexed for range queries and analytics

Coupon

Discount codes that can be applied at cart or checkout.

Field	Type	Constraint	Description
_id	ObjectId	PK	Auto-generated document identifier
code	String	Required, Unique	Coupon code string (e.g. SAVE20)
discountType	String	Required	Enum: percentage fixed
discountValue	Number	Required	Discount amount or percentage value

minOrderAmount	Number	Optional	Minimum cart total required to apply the coupon
usageLimit	Number	Optional	Maximum number of total redemptions allowed
usageCount	Number	Optional	Running count of how many times the coupon has been used
isActive	Boolean	Required	Whether the coupon is enabled
expiresAt	Date	Optional	Coupon expiry date and time